ENEL 573 – Practice Questions

Problem 1

a) The minimum distance from the Earth to Mars is about 54.6 million kilometers. One computer on the Earth and another on Mars communicate by using the stop-and-wait protocol. If the computer on the Earth sends 1.5 Kilobyte DATA frames to the computer on Mars at a transmission speed of 10 Megabits/second, compute the transmission time of a DATA frame. Assume that wireless signal propagates in free space at the speed of 3×10^8 meters/sec.

Solution:

1.5 Kilobytes = 1.5 x 8 Kilobits = 12 Kilobits
To transmit 12 Kilobits, the time needed is = 12×10^3 bits/(10×10^6 bits/ sec.) = 1.2 ms.
The transmission time = 1.2 ms. = 0.0012 sec.

b) In the networking scenario described in part a), compute the minimum total time needed to completely copy one DATA frame from the computer on the Earth to the computer on Mars. Ignore queuing delay and processing delay.

Solution:

Total time = Transmission delay + Propagation delay
The transmission delay can be found in part a).
Distance between the Earth and Mars = $54.6 \times 10^6 \times 10^3$ m = 5.46×10^{10} m
Signal propagation speed = 5.46×10^{10} m/sec
Propagation delay = distance/propagation speed = 5.46×10^{10} m/ (3×10^8 m/sec)
= 5.46 x 100/3 sec = 182 sec.

1.2 ms << 182 sec.
Therefore, the time to copy a 1.5 Kilobyte data packet from the Earth to Mars =~ 182 sec.

c) In the networking scenario described in part a), assume that the length of ACK frames is negligible compared to the length of DATA frames and you replace the stop-and-wait protocol with a pipelined protocol. Calculate the maximum number of DATA frames that can be transmitted by a computer before waiting for an ACK.

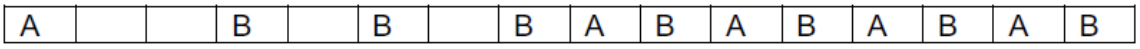
Solution:

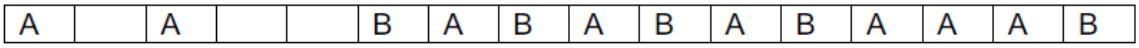
The cycle time = Transmission delay + 2 x Propagation delay = 0.0012 sec + 2 x 182 sec = 364.0012 sec
of DATA frames transmitted in one cycle time = 364.0012 sec/1.2 msec = 303,334.

A computer can send 303,334 DATA frames in a pipelined protocol before stopping for an ACK.

d) Assume that a hardware-software tool can capture data frames being transmitted over a variety of media and know the sender's identity. Two communication scenarios have been shown

in what follows, time passes from left to right, and A and B represent the sender's identity in frames transmitted in those periods. An empty cell indicates the absence of a data frame. For simplicity, assume that all the cells are of identical durations. Identify the most probable multiplexing techniques that might have been used in the two scenarios.





Solution:

The first scenario: TDM or statistical multiplexing
The second scenario: Statistical multiplexing, because of the AAA toward the end.

e) What is meant by "circuit-like" behavior of computer networks?
Solution:

Circuit-like behaviour means achieving guaranteed performance, because of resource reservation along the path (circuit):

(i) The data transfer rate is constant.
(ii) The end-to-end delay is constant,

f) In what protocol from the set {pure Aloha, slotted Aloha, CSMA/CD} more packets will collide than the others? Assume that the networks have the same number of nodes attempting to transmit data in the same way.
Solution:
Pure Aloha

Problem 2

a) Consider two nodes (computers or routers) that have been connected by means of a single optical fibre link of length 2500 kilometers. Assume that optical signal propagates over the link at the speed of 2.5x108 m/s (metre/sec), and the link speed is 2 Mbps. If computer A is sending a frame of length 1,000 bytes to B, how long does it take for the frame to be completely copied from A to B?
Solution:

Basic idea: Time to completely copy a frame, Tc = Tp (time to propagate) + Tx (time to transmit)

Tp = Distance/ speed of signal = 2500 Km/(2.5×10^8 m/s) = 2500×10^3 m /(2.5×10^8 m/s)
= 2.5×10^6 m /(2.5×10^8 m/s) = 10^{-2} s = 10 ms.

Tx = Length of a frame / Link speed = 1000 bytes / 2 Mbps = 1000 x 8 bits/ (2×10^6 bps)
= 8×10^3 b/(2×10^6 b/s) = 8×10^3 /(2×10^6) s = 4×10^{-3} s = 4 ms.

Therefore, Tc = 10 ms + 4 ms = 14 ms.
Answer: It takes **14 ms** to completely copy a frame of 1000 bytes from A to B.

b) Recall that with the CSMA/CD protocol, the adapter waits K× 512 bit times after a collision, where K is drawn randomly. For K = 100, how long does the adapter wait (in seconds) before starting to sense the medium for a 100 Mbps broadcast channel?
Solution:

Basic idea: The wait time, Tw = K x 512 bit times.
For a 100 Mbps broadcast channel, 100×10^6 (= 10^8) bits are transmitted in 1 second.
Therefore, 1 bit time = 1/(10^8) seconds = 10^{-8} s.

Tw = 100 x 512 x 10^{-8} s. = 512×10^{-6} s.

Answer: The adapter waits for **512×10^{-6} s** before starting to sense the medium.

c) Suppose two nodes (computers or routers) A and B are on the same 10 Mbps Ethernet broadcast link. Node A detected a collision after transmitting 60 Bytes from a frame, and the corresponding delay in detecting collision was the longest among millions of its transmissions. What might be the propagation delay (in seconds) between the two nodes? Justify your answer (Hint: How long maximumly does it take for CSMA/CD to detect the collision?).
Solution:
Basic idea: The maximum number of bits transmitted by a node before detecting collision is the number of bits that can be transmitted within 2 x Tp seconds, where Tp is the propagation time between the two nodes.

Node A transmitted 60 bytes (= 60 x 8 = 480 bits) from a frame before detecting collision.
In 2 x Tp seconds, node A transmitted 480 bits.
In Tp second, node A transmitted 240 (= 480/2) bits.
Therefore, the propagation delay between the two nodes, Tp = 240 bit times.
1 bit time in 10Mbps Ethernet is 1/($10*10^6$) = 10^{-7}s
So, Tp = $240*10^{-7}$s

Problem 3
a) Consider the 5-bit generator, G=10011. If a receiver receives a data frame represented by the bit string 1010110 with CRC bits in it, was the frame received correctly? Justify your answer.

Basic idea: Divide the frame (= 1010110), including the CRC, by G (= 10011). If the remainder is 0000, then the frame has been received correctly. Otherwise, it has been incorrectly received.

```
Frame:    1010110
G:        10011
----------------------------------------------------------
          0011010   (The bit in the frame above the MSB of G was 1 so XOR frame and G.
           10011    (Shift G to right by 1 position.)
----------------------------------------------------------
          0011010   (The bit in the "frame" above the MSB of G was 0 so do NOT XOR frame and G.
            10011   (Shift G to right by 1 position.)
----------------------------------------------------------
          0001001   (The bit in the frame above the MSB of G was 1 so XOR frame and G.)
```

The computation stops now, because G has been shifted to the right-hand end of the "frame."

The remainder of the division is **1001**.

Answer: No, the frame was not received correctly by the receiver.

b) Clearly explain the reason for the **slotted Aloha** protocol giving better throughput than the pure Aloha protocol.

Basic idea: Explain that the probability of collision of a frame in slotted Aloha is less than the probability of collision in pure Aloha.

Answer: Let Tf be the time to transmit a frame.

In the slotted Aloha protocol, nodes are constrained to transmit **at the beginning of each slot**, whereas in pure Aloha, nodes can transmit **any time**. Therefore, in pure Aloha, a frame (say, X) can collide with another transmission (say, Y, from a different node) that started within Tf seconds before X. Similarly, in pure Aloha, a frame (say, X) can collide with another transmission (say, Z, from a different node) that started within Tf seconds of X's transmission.

On the other hand, in the slotted Aloha protocol, Y and Z cannot collide with X.

Therefore, the slotted Aloha protocol gives better performance than the pure Aloha.

(You could explain it by means of a figure as well, as done in class.)

c) Show the operational behavior of a receiver running the **pure Aloha** protocol.

Answer: (You could draw a flow-chart for this.)

(i) The receiver (node B) waits for a frame, F, from another node (A).

(ii) Check for bit errors in F.

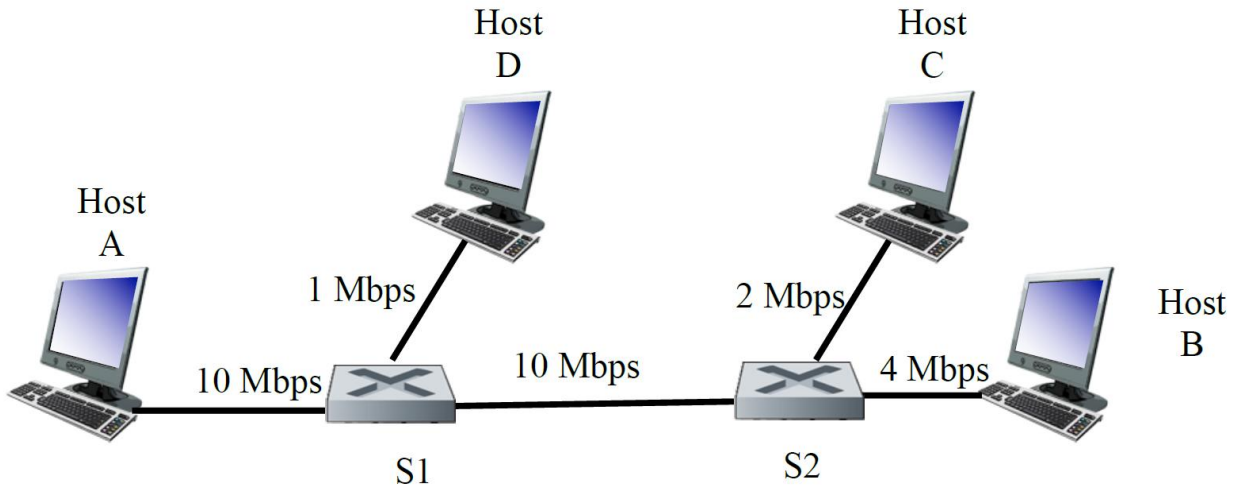
(iii) If there is no bit error in F,
then extract DATA from F and deliver DATA to the upper layer, and **send** an ACK (with error checking code) to A using the pure Aloha protocol. (There is no need to show the Sender's behavior.)
else do nothing (Reject F.)

(iv) Go back to step (i).

Note: Even if you did not do error checking, you will get full credit.

Problem 4

In the packet-switched network shown in the figure below, each link has its speed. S1 and S2 are link-layer switches. Ignore the processing, propagation and queuing delays at all the switches. Host A will broadcast a file of size 100 KBytes. Host A has a fixed packet size of 1 KBytes. Consider that the transmission of packet 1 will start at **time 0**. Answer the following questions 1) - 6):



1) When will Host D completely receive the first packet?
2) When will Host C completely receive the first packet?
3) When will Host B completely receive the first packet?
4) When will Host C completely receive the file?
5) When will Host B completely receive the file?
6) Which Host out of Host B and Host C will have better average throughput. Justify your answer.

Solution:

1) When will Host D completely receive the first packet?

$$T_{D-1st-packet} = d_{trans-10M} + d_{trans-1M}$$

$$T_{D-1st-packet} = \frac{1000 \times 8}{10 \times 10^6} + \frac{1000 \times 8}{10^6}$$

$$T_{D-1st-packet} = 0.8 + 8$$

$$T_{D-1st-packet} = 8.8\ msec$$

2) When will Host C completely receive the first packet?

$$T_{C-1st-packet} = d_{trans-10M} + d_{trans-10M} + d_{trans-2M}$$

$$T_{C-1st-packet} = \frac{1000 \times 8}{10 \times 10^6} + \frac{1000 \times 8}{10 \times 10^6} + \frac{1000 \times 8}{2 \times 10^6}$$

$$T_{C-1st-packet} = 0.8 + 0.8 + 4$$

$$T_{C-1st-packet} = 5.6\ msec$$

3) When will Host B completely receive the first packet?

$$T_{B-1st-packet} = d_{trans-10M} + d_{trans-10M} + d_{trans-4M}$$

$$T_{B-1st-packet} = \frac{1000 \times 8}{10 \times 10^6} + \frac{1000 \times 8}{10 \times 10^6} + \frac{1000 \times 8}{4 \times 10^6}$$

$$T_{B-1st-packet} = 0.8 + 0.8 + 2$$

$$T_{B-1st-packet} = 3.6\ msec$$

4) When will Host C completely receive the file?

$$Number\ of\ packets = \frac{100K}{1K} = 100\ packets$$

$$T_{C-complete} = T_{C-1st-packet} + (number\ of\ packets - 1) \times d_{trans-2M}$$

$$T_{C-complete} = 5.6 + (100 - 1) \times 4$$

$$T_{C-complete} = 401.6\ msec$$

5) When will Host B completely receive the file?

$$T_{B-complete} = T_{B-1st-packet} + (number\ of\ packets - 1) \times d_{trans-4M}$$

$$T_{C-complete} = 3.6 + (100 - 1) \times 2$$

$$T_{C-complete} = 201.6\ msec$$

6) Which Host out of Host B and Host C will have better average throughput. Justify your answer.

$$Throughput_C = \frac{100\ Kbytes}{T_{C-complete}} = \frac{100K}{401.6 \times 10^{-3}} = 249\ Kbytes/sec$$

$$Throughput_B = \frac{100\ Kbytes}{T_{B-complete}} = \frac{100K}{201.6 \times 10^{-3}} = 496.04\ Kbytes/sec$$

Host B has higher throughput than that of C (almost double) as the last link in the connection is bottleneck link (4 Mbps vs 2 Mbps).

Problem 5

Assume that the number of packets *n* (*n* packets corresponds to *n* transmitting nodes) in a **slotted Aloha** system at a given time is a Poisson random variable with mean $\bar{n} \geq 1$. Suppose each packet is independently transmitted in the next slot with probability $1/\bar{n}$.

(a) Find the probability that the slot is idle.

(b) Find the probability that the slot is successful.

(a) Given k packets in the system, with each packet independently transmitted in a slot with probability $1/\overline{n}$, the probability of an idle slot, $P\{I \mid k\}$ is $(1-\frac{1}{\overline{n}})^k$. The joint probability of an idle slot and k packets in the system is then

$$P\{I,k\} = P\{I \mid k\}P\{k\} = (1-\frac{1}{\overline{n}})^k \cdot \frac{e^{-\overline{n}}\overline{n}^k}{k!} = \frac{e^{-\overline{n}}(\overline{n}-1)^k}{k!}$$

$$P\{I\} = \sum_{k=0}^{\infty} P\{I,k\} = \sum_{k=0}^{\infty} \frac{e^{-\overline{n}}(\overline{n}-1)^k}{k!} = \frac{1}{e}$$

(c) Similarly as part (a), we can first find the joint probability of success and k in the system, namely $P\{S,k\}$, then calculate the successful probability $P\{S\}$.

$$P\{S,k\} = P\{S|k\}P\{k\} = k(1-\frac{1}{\overline{n}})^{k-1}\frac{1}{\overline{n}} \cdot \frac{e^{-\overline{n}}\overline{n}^k}{k!} = \frac{e^{-\bar{n}}(\bar{n}-1)^{k-1}}{(k-1)!}$$

$$P\{S\} = \sum_{k=1}^{\infty} P\{S,k\} = \sum_{k=1}^{\infty} \frac{e^{-\overline{n}(\overline{n}-1)^{k-1}}}{(k-1)!} = \frac{1}{e}$$